BMS_MASTER

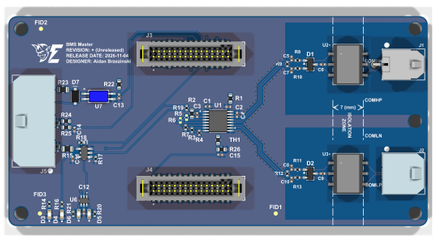
Variant: PRELIMINARY

2026-11-04

Rev + (Unreleased)

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

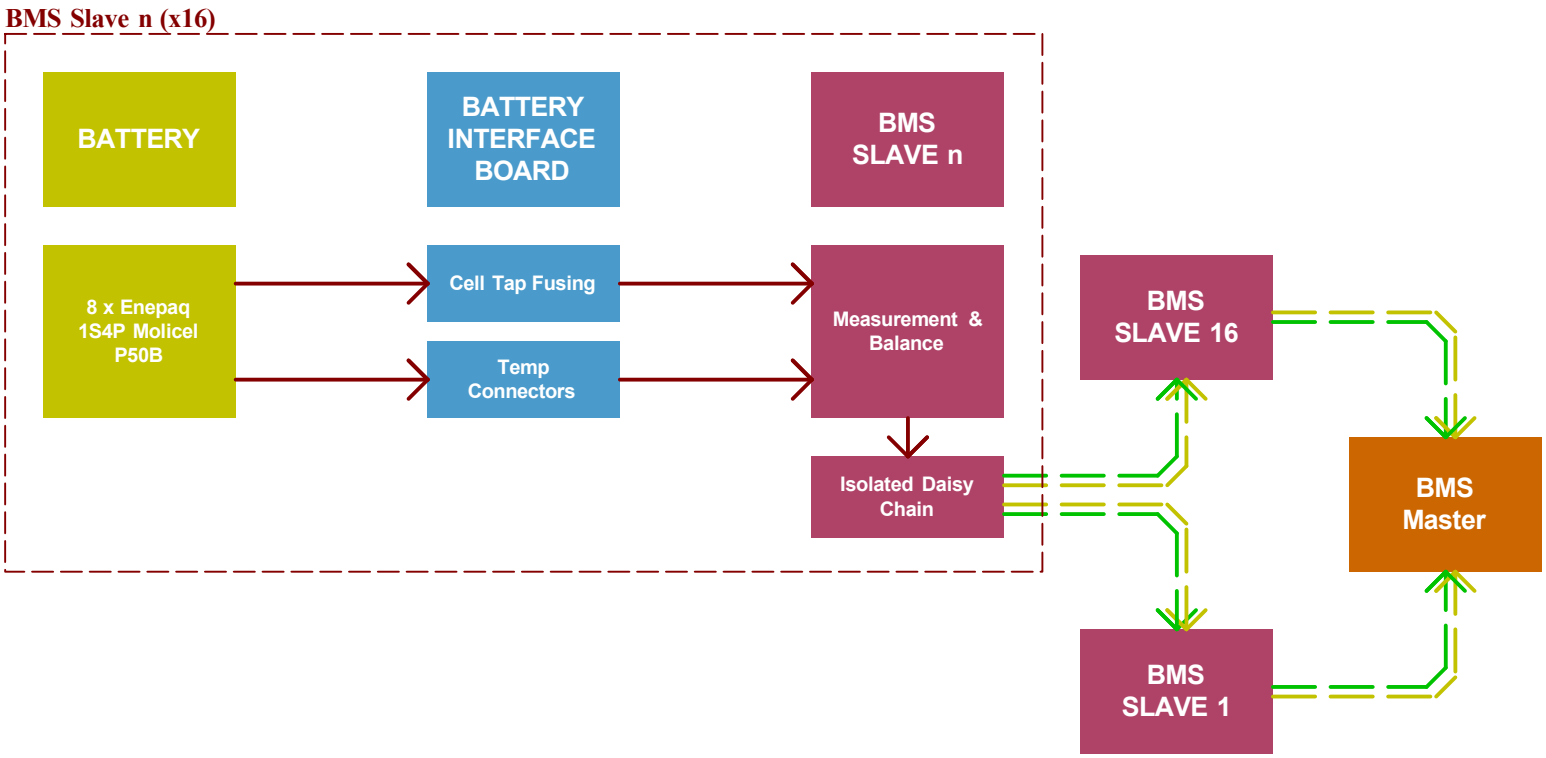
DESIGN NOTE: Example text for cautionary design notes.

DESIGN NOTE: Example text for critical design notes.

LAYOUT NOTE: Example text for critical layout guidelines.

Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 45e7b92	
	Board Name: BMS_MASTER		Project Name: BMS Master		
Sheet Title:	File Name: bms_master.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)	
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 10	

[2] Block Diagram

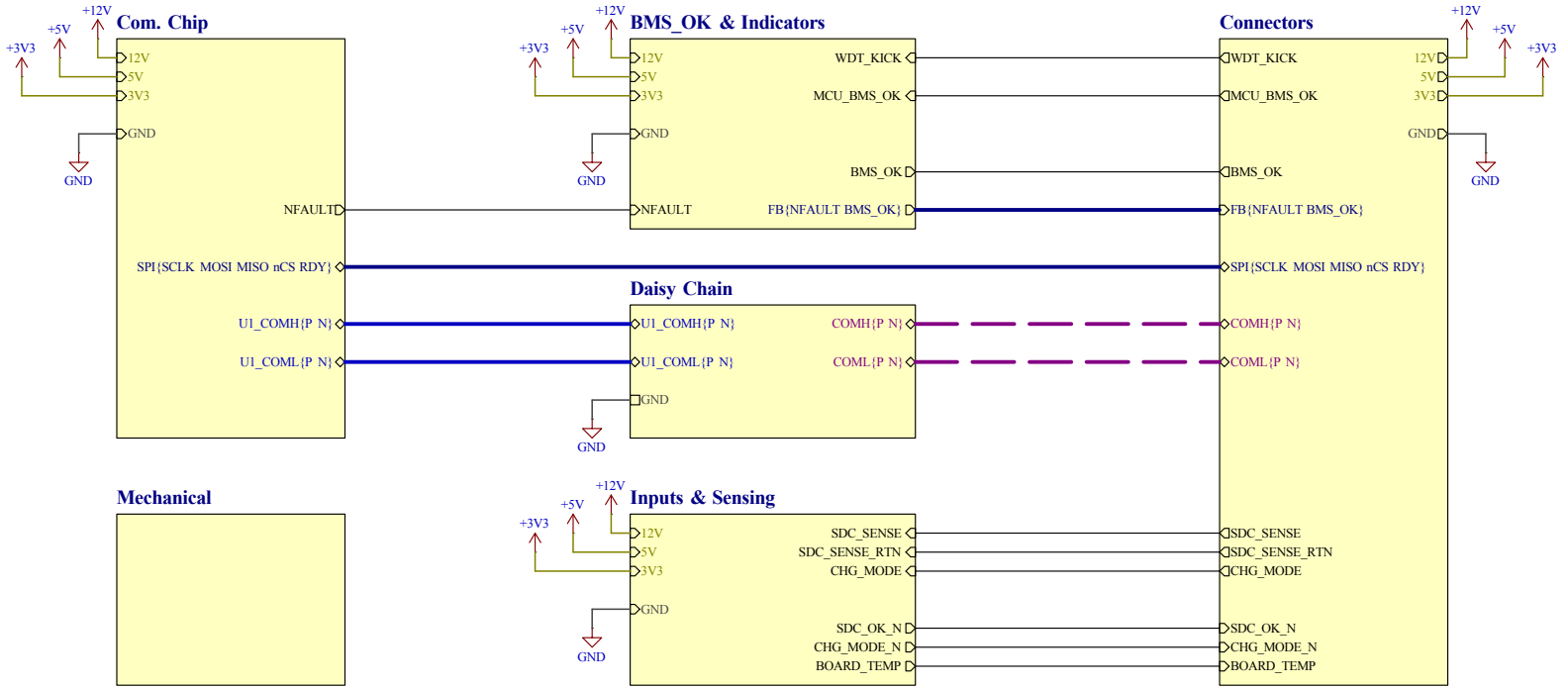


Target specifications:

Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

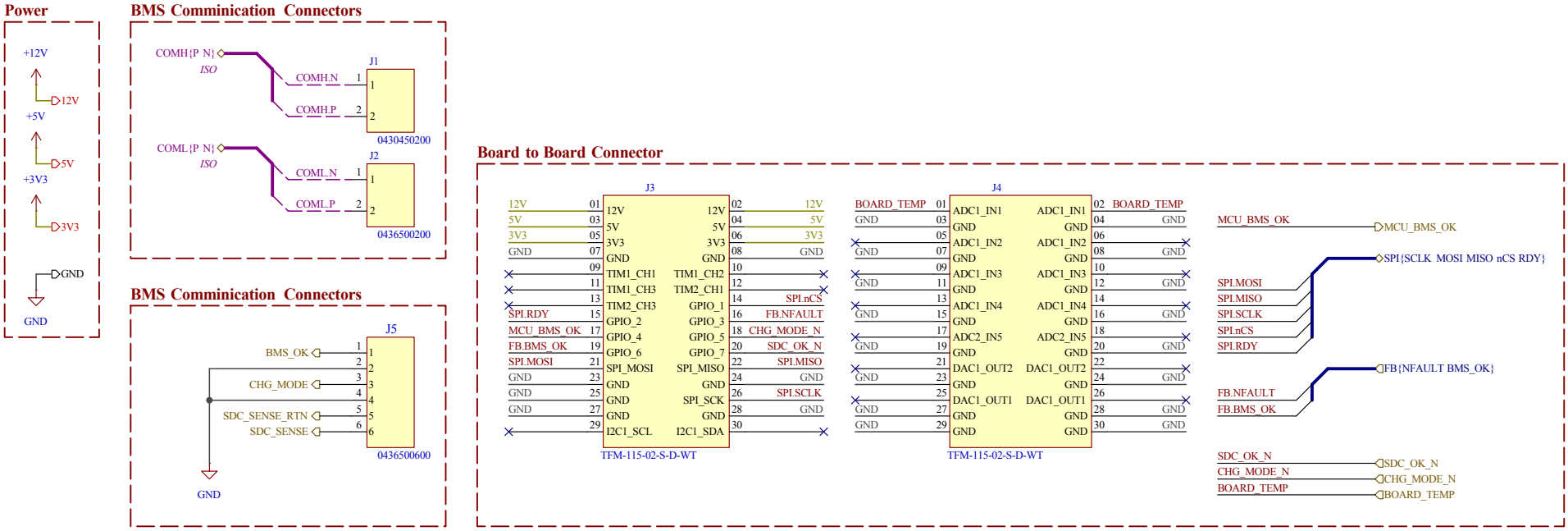
Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 45e7b92
	Board Name: BMS_MASTER		Project Name: BMS Master	
Sheet Title: Block Diagram	File Name: Block Diagram.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)
Sheet Path: /Block Diagram/		Reviewer:	Size: A4	Sheet: 2 of 10

[3] Project Architecture

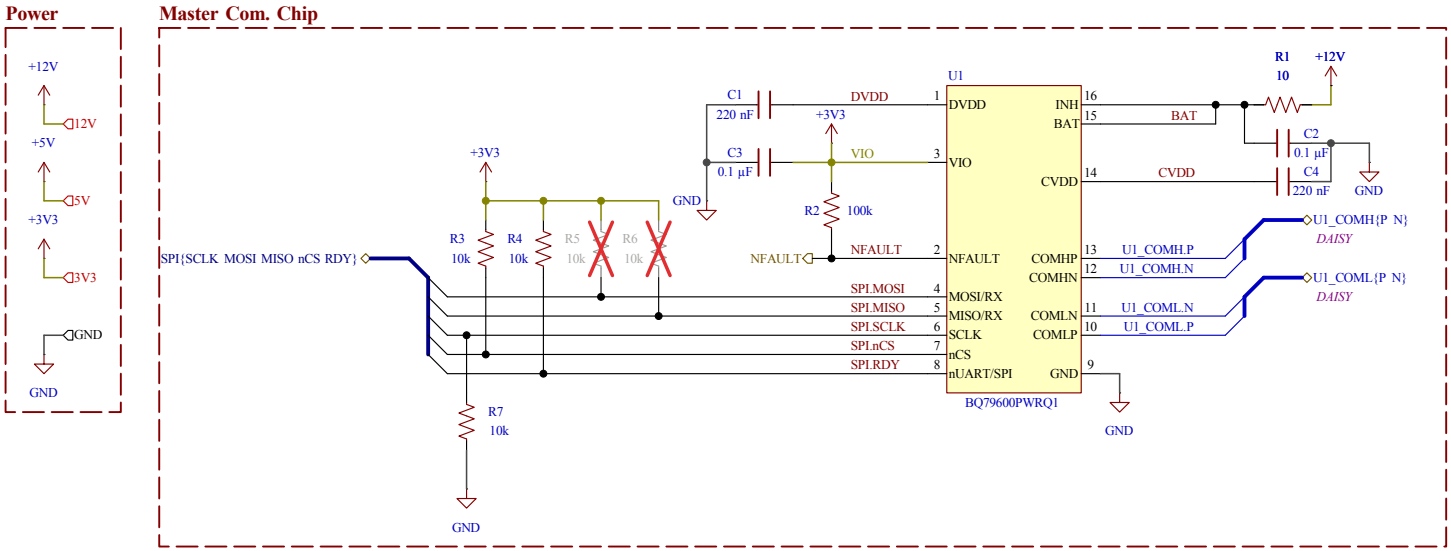


Comments:	Company: **Blue Devil Motor Sports**		Variant: PRELIMINARY	Git Hash: 45e7b92
	Board Name: **BMS_MASTER**		Project Name: **BMS Master**	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: **A4**	Sheet: **3** of **10**

[4] Connectors


Power
+12V
12V
+5V
5V
+3V3
3V3
GND
GND
BMS Comminication Connectors
COMH{P N}
ISO
COMH_N
COMH_P
J1
1
2
0430450200
COML{P N}
ISO
COML_N
COML_P
J2
1
2
0436500200
BMS Comminication Connectors
J5
BMS_OK
CHG_MODE
SDC_SENSE_RTN
SDC_SENSE
1
2
3
4
5
6
0436500600
GND
Board to Board Connector
J3
12V 01
5V 03
3V3 05
GND 07
09
11
13
SPI.RDY 15
MCU_BMS_OK 17
FB.BMS_OK 19
SPI.MOSI 21
GND 23
GND 25
GND 27
29
12V 02
5V 04
3V3 06
GND 08
10
12
SPI.nCS 14
FB.NFAULT 16
CHG_MODE_N 18
SDC_OK_N 20
SPI.MISO 22
GND 24
SPI.SCLK 26
GND 28
30
12V
5V
3V3
GND
TIM1_CH1
TIM1_CH3
TIM2_CH3
GPIO_2
GPIO_4
GPIO_6
SPI_MOSI
GND
GND
GND
I2C1_SCL
TIM1_CH2
TIM2_CH1
GPIO_1
GPIO_3
GPIO_5
GPIO_7
SPI_MISO
GND
SPI_SCK
GND
I2C1_SDA
TFM-115-02-S-D-WT
J4
BOARD_TEMP 01
GND 03
05
GND 07
09
GND 11
13
GND 15
17
GND 19
21
GND 23
25
GND 27
GND 29
ADC1_IN1 02
GND 04
ADC1_IN2 06
GND 08
ADC1_IN3 10
GND 12
ADC1_IN4 14
GND 16
ADC2_IN5 18
GND 20
DAC1_OUT2 22
GND 24
DAC1_OUT1 26
GND 28
GND 30
ADC1_IN1
GND
ADC1_IN2
GND
ADC1_IN3
GND
ADC1_IN4
GND
ADC2_IN5
GND
DAC1_OUT2
GND
DAC1_OUT1
GND
GND
BOARD_TEMP
GND
GND
GND

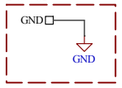
[5] Com. Chip



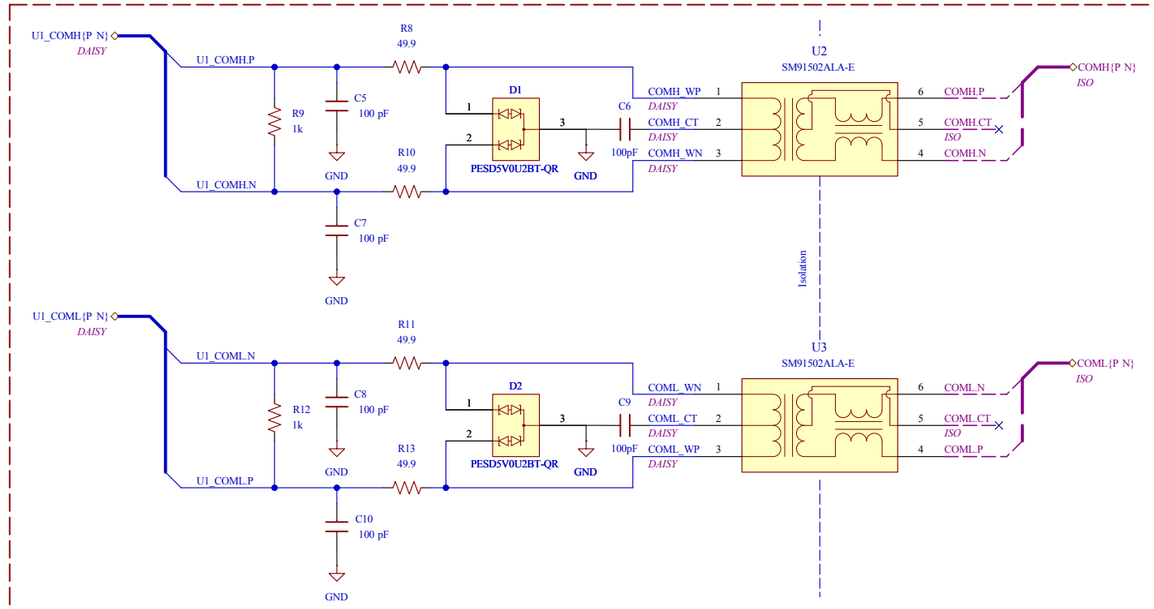
Comments:	Company: **Blue Devil Motor Sports**		Variant: PRELIMINARY	Git Hash: 45e7b92
	Board Name: **BMS_MASTER**		Project Name: **BMS Master**	
Sheet Title: Com. Chip	File Name: com_chip.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Com. Chip/		Reviewer:	Size: **A4**	Sheet: **5** of **10**

[6] Daisy Chain

Power



Communication Isolation



DESIGN NOTE: DAISY CHAIN ISOLATION

1. Transformer isolation per SLUSE81F Table 8-5 / Fig 8-9: R_TERM 1 kΩ, R_SERIES 49.9 Ω, C_BYPASS 220 pF/50 V C0G, centre-tap cap 100 pF to GND, TVS PESD1CAN on chip side.
2. Cable-side centre taps (U2-5, U3-5) are left NC per Fig 8-9.
3. Stack device: RX tied to CVDD, TX and NFAULT left NC (SLUSE81F 8.2.1.2.6).
4. U3 is drawn with chip-side pins 1/3 opposite to U2. Polarity is preserved end to end (P->P, N->N).
5. Ring architecture: COMH (J2) goes up the stack, COML (J1) goes down. J1 = 43650-0200 single-row, J2 = 43045-0200 dual-row so a cable cannot connect COMH to COMH.
6. Use the same transformer PN on both ends of every link, including the master (TI SLVAEP4 5.1).

LAYOUT NOTE: DAISY CHAIN (TI SLUSE81F 10.1.4, Fig 10-3)

1. COMH and COML pairs: short, straight, same layer, parallel, length matched. No vias in the pairs.
2. R_TERM and C_BYPASS close to U1 pins 40-43.
3. Solid GND under the chip-side parts (R, C, TVS) is required for the bypass and TVS returns. Keep unrelated traces out of this area, and never route a COM pair across a plane seam.
4. No copper on ANY layer under the transformer bodies or between the transformer cable side and J1/J2.
5. Transformer barrier: 7.11 mm pad gap meets EV.7.5.7 300-600 V coated creepage (4.0 mm) and clearance (3.0 mm). Keep >= 4.0 mm on all layers. No plated holes, vias or pours inside the isolation zone.
6. Place each transformer next to its connector.
7. TP11-TP14: bare SMD pads on the trace, no stubs.

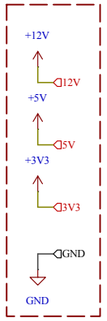
CAUTION: ISOLATION SPACING (FSAE 2027 EV.8.3.2, EV.7.5.7)

1. EV.8.3.2: the BMS must have galvanic isolation at every Module-to-Module boundary. U2/U3 provide it. All slaves are identical, so every barrier is designed for the worst case: full pack voltage, <128S x 4.2 V = 537.6 V> max (EV.3.3.2 limit: 600 V DC).
2. EV.7.5.7 spacing, 300-600 V DC row:
Clearance (through air): 3.0 mm
Creepage, general (uncoated): 20.0 mm
Creepage, under conformal coating: 4.0 mm
3. This board IS conformally coated, so the 4.0 mm creepage and 3.0 mm clearance apply. Uncoated it would NOT comply (7.11 mm < 20 mm).
4. Rule applies on the same layer AND between layers. Keep >= 4.0 mm on every layer between chip-side copper (inner GND pours, vias, TVS / CT-cap pads) and cable-side copper (U2/U3 pins 4-6, J1/J2). Actual: 7.11 mm transformer pad gap (SM91502ALA land pattern).
5. No plated holes, vias or copper pours inside the isolation zone.
6. Mark the zone on top AND bottom silkscreen (boundary lines + "ISOLATION ZONE"), per EV.7.5.6 / EV.7.5.7a.
7. U2/U3 isolation rating and the daisy-chain cable insulation must be rated >= max TS voltage (EV.5.2.1, EV.7.3.2).
8. Keep a spare coated board or photos for inspection (EV.7.5.8).

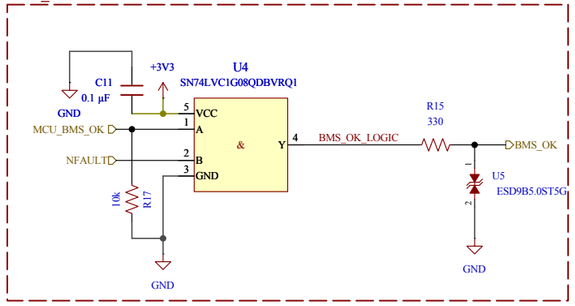
Comments:	Company: **Blue Devil Motor Sports**		Variant: PRELIMINARY	Git Hash: 45e7b92
	Board Name: **BMS_MASTER**		Project Name: **BMS Master**	
Sheet Title: Daisy Chain	File Name: daisy_chain.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Daisy Chain/		Reviewer:	Size: **A3**	Sheet: **6** of **10**

[7] BMS_OK & Indicators

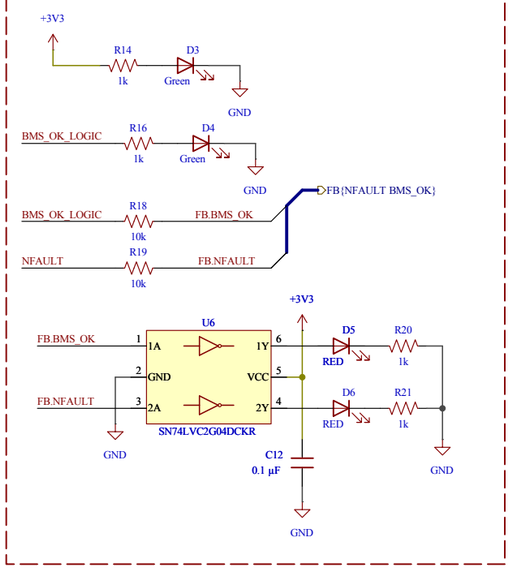
Power



BMS_OK AND Gate



Indcator Leds & Feedback



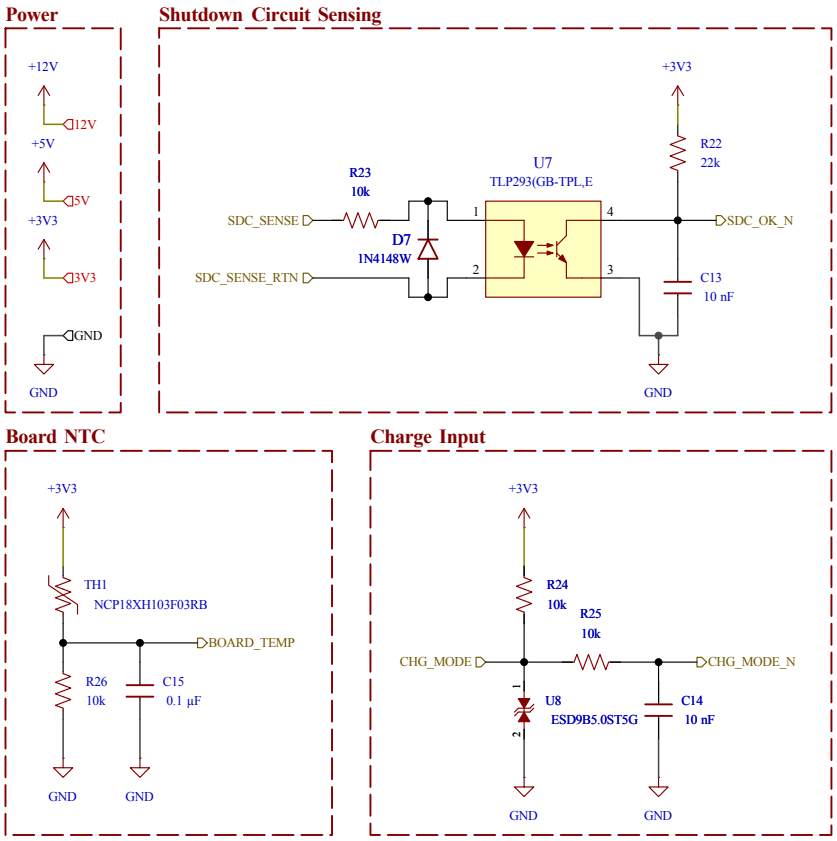
DESIGN NOTE: BMS_OK LOGIC
BMS_OK = NFAULT AND MCU_BMS_OK AND WDT_OK. Any input low -> no opto current -> BMS relay opens.
- NFAULT: BQ79600 open-drain fault, 100k pull-up on Com. Chip page (SLUSDS1A p.3).
- MCU_BMS_OK: high only after a fresh valid stack read; 10k pull-down holds it low in reset.
- WDT_OK: TPS3430 window watchdog, CWD=10k, SET0=0, SET1=1 -> window 10.35-165.8 ms
  worst case (SBVS366A p.18). Kick on falling edge every 100 ms.
- U4/U5 outputs go high-Z when unpowered (Ioff), so loss of 3V3 = BMS_OK off.
- R_OUT 330R: ~6 mA nominal, ~3 mA worst case into the opto.
- Readbacks/LEDs are behind 10k so a diagnostic fault can't force a safety net high.

LAYOUT NOTE: 100 nF caps at each VCC/VDD pin. TPS3430 pad to GND with vias. Keep WDT_CRST short (high-impedance node). R_OUT near J5.

CAUTION: WATCHDOG LATCH
U5 holds CRST low once WDO trips, so WDO stays low until 3V3 drops below 1.6 V (TPS3430 Design 3, SBVS366A p.23-24). A watchdog trip needs a GLV power cycle AND the shutdown-board BMS reset. CRST 4.7 nF -> ~15.5 ms startup grace; firmware must start kicking before tRST + tWDU(min).

Comments:	Company: **Blue Devil Motor Sports**		Variant: PRELIMINARY	Git Hash: 45e7b92
	Board Name: **BMS_MASTER**		Project Name: **BMS Master**	
Sheet Title: BMS_OK & Indicators	File Name: bms_ok_and_indicators.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/BMS_OK & Indicators/		Reviewer:	Size: **A3**	Sheet: **7** of **10**

[8] Inputs & Sensing



Comments:	Company: **Blue Devil Motor Sports**		Variant: PRELIMINARY	Git Hash: 45e7b92
	Board Name: **BMS_MASTER**		Project Name: **BMS Master**	
Sheet Title: Inputs & Sensing	File Name: inputs_and_sensing.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Inputs & Sensing/		Reviewer:	Size: **A4**	Sheet: **8** of **10**

[9] Mechanical

Mounting Holes

- H1 MountingHole
- H2 MountingHole
- H3 MountingHole
- H4 MountingHole
- H5 MountingHole

Fiducials

- FID1 Fiducial
- FID2 Fiducial
- FID3 Fiducial

Silkscreen

BMD Logo

SAE Electric logo

Comments:	Company: Blue Devil Motor Sports		Variant: PRELIMINARY	Git Hash: 45e7b92
	Board Name: **BMS_MASTER**		Project Name: **BMS Master**	
Sheet Title: Mechanical	File Name: mechanical.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Mechanical/		Reviewer:	Size: **A4**	Sheet: **9** of **10**

[10] Revision History

Comments:	Company: **Blue Devil Motor Sports**		Variant: PRELIMINARY	Git Hash: 45e7b92	
	Board Name: **BMS_MASTER**		Project Name: **BMS Master**		
Sheet Title: Revision History	File Name: Revision History.kicad_sch	Designer: Aidan Brzezinski	Date: 2026-10-20	Revision: + (Unreleased)	
Sheet Path: /Revision History/		Reviewer:	Size: **A4**	Sheet: **10** of **10**	